__/__/____

Product of Array Except Self

A = | 1 | 2 | 3 | 4 |

→ Essentially, at every index we have to store what is the multiplication of all the elements, except the current element of the index

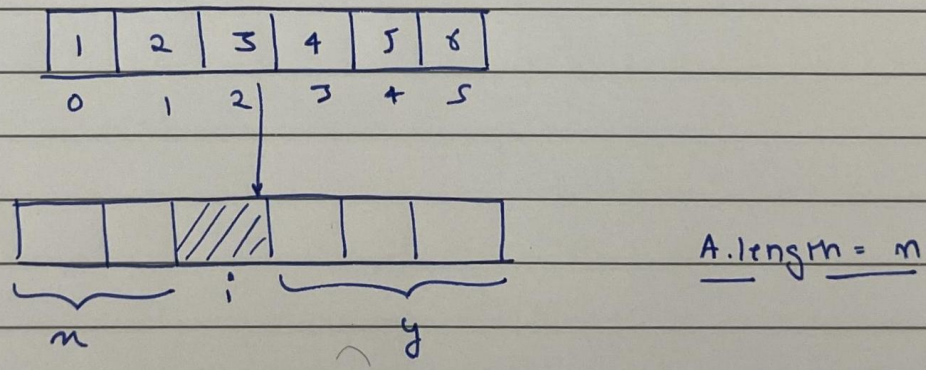
→ This problem is a 'no-brainer' if we were simply allowed to calculate the multiplicat^{n} of all the elements beforehand, & for each array element, divide the multiplicat^{n} result with the current array element.

→ However, the problem has strictly discouraged the use of the 'division operator' i.e '/'

→ And this exactly what makes the problem interesting!

Approach (T.C. - O(N) & S.C. - O(N))

→ One thing is certain, we definitely need some kind of pre-computation



A.length = n

A[i] = x x y, x is the multiplicat^{n} of all elements before i
y is the multiplicat^{n} of all elements after i

→ For 'x': we can carry forward the multiplicat^{n} of each element as we progress, but what about the elements present ahead?

→ For 'y': I pre-computed a suffix array, which stores the multiplicat^{n} of all elements, just ahead of each index!

∴, answer[i] = x x suffix[i]